

ALL NCERT

BIOLOGY

SCIENTISTS

& DATES

ONE SHOT 



All NCERT Biology Scientists and Dates

IN ONE SHOT



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Ma'am



THE LIVING WORLD

①

❖ Ernst Mayr (1904-2004) :

- Darwin of 20th century (PYQ)
- Triple crown of biology
- Gave currently accepted definition of Biological species

❖ This system of providing a name with two components is called Binomial nomenclature. This naming system given by Carolus Linnaeus is being practised by biologists all over the world.

❖ Linnaeus used Systema Naturae as title of his publication.

② → Father of Taxonomy



BIOLOGICAL CLASSIFICATION

③ Aristotle earliest to attempt a more scientific basis for classification.

④ → Father of Biology

In Linnaeus' time a Two Kingdom system of classification with Plantae and Animalia kingdoms was developed

⑤ R.H. Whittaker (1969) proposed a Five Kingdom Classification.

1969



⑥

V. Gmp.

Dmitri Ivanowsky (1892) recognised certain microbes as causal organism of the mosaic disease of tobacco. These were found to be smaller than bacteria because they passed through bacteria-proof filters.

⑦

M.W. Beijerinck (1898) demonstrated that extract of infected plants of tobacco could cause infection in healthy plants and called fluid as Contagium vivum fluidum. He called these new organisms as 'Viruses' (venom/poisonous)

⑧

W.M. Stanley (1935) viruses could be crystallised and crystals consist largely of proteins.



9
Viroids : In 1971, T.O. Diener discovered



PLANT KINGDOM





The earliest systems of classification used only gross superficial morphological characters such as habit, colour, number and shape of leaves, etc. They were based mainly on vegetative characters or on the androecium structure (system given by Linnaeus). Such systems were artificial;

10



As against this, natural classification systems developed, which were based on natural affinities among the organisms and consider, not only the external features, but also internal features, like ultrastructure, anatomy, embryology and phytochemistry. Such a classification for flowering plants was given by George Bentham and Joseph Dalton Hooker.



ANIMAL KINGDOM



MORPHOLOGY OF FLOWERING PLANTS

ANATOMY OF FLOWERING PLANTS

Structural organization in animals



12

Katherine ESAU, The anatomy of seed plants (referred as Webster's of plant biology)
published in 1960.



CELL: THE UNIT OF LIFE

13

G.N.Ramchandran, Protein structure, Triple helical structure of collagen

Ramchandran plot – analysis of conformations of proteins.

PYQ





14

In 1838, Matthias Schleiden, a German botanist, examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant.

15

22/11 = Dog

Theodore Schwann (1839), a British Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'. He also concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells.



On the basis of this, **Schwann** proposed the hypothesis that the bodies of animals and plants are composed of cells and products of cells.

Schleiden and Schwann together formulated the cell theory.

¹⁶
Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (Omnis cellula-e cellula). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape.



Bacteria can be classified into two groups on the basis of the differences in the cell envelopes and the manner in which they respond to the staining procedure developed by Gram viz., those that take up the gram stain are Gram positive and the others that do not are called Gram negative bacteria.

17





18

An improved model of the structure of cell membrane was proposed by Singer and Nicolson (1972) widely accepted as fluid mosaic model





19



Camillo Golgi (1898) first observed **densely stained reticular structures near the nucleus**.
These were later named Golgi bodies after him.





20

Ribosomes are the **granular structures** first observed under the **electron microscope** as dense particles **by George Palade (1953)**.





21

Nucleus as a cell organelle was first described by Robert Brown as early as 1831.

Later the material of the nucleus stained by the basic dyes was given the name chromatin by Flemming.

22





BIOMOLECULES

Secondary structures exhibited by DNA is the famous Watson-Crick model

23

Helix model





Cell cycle and cell division





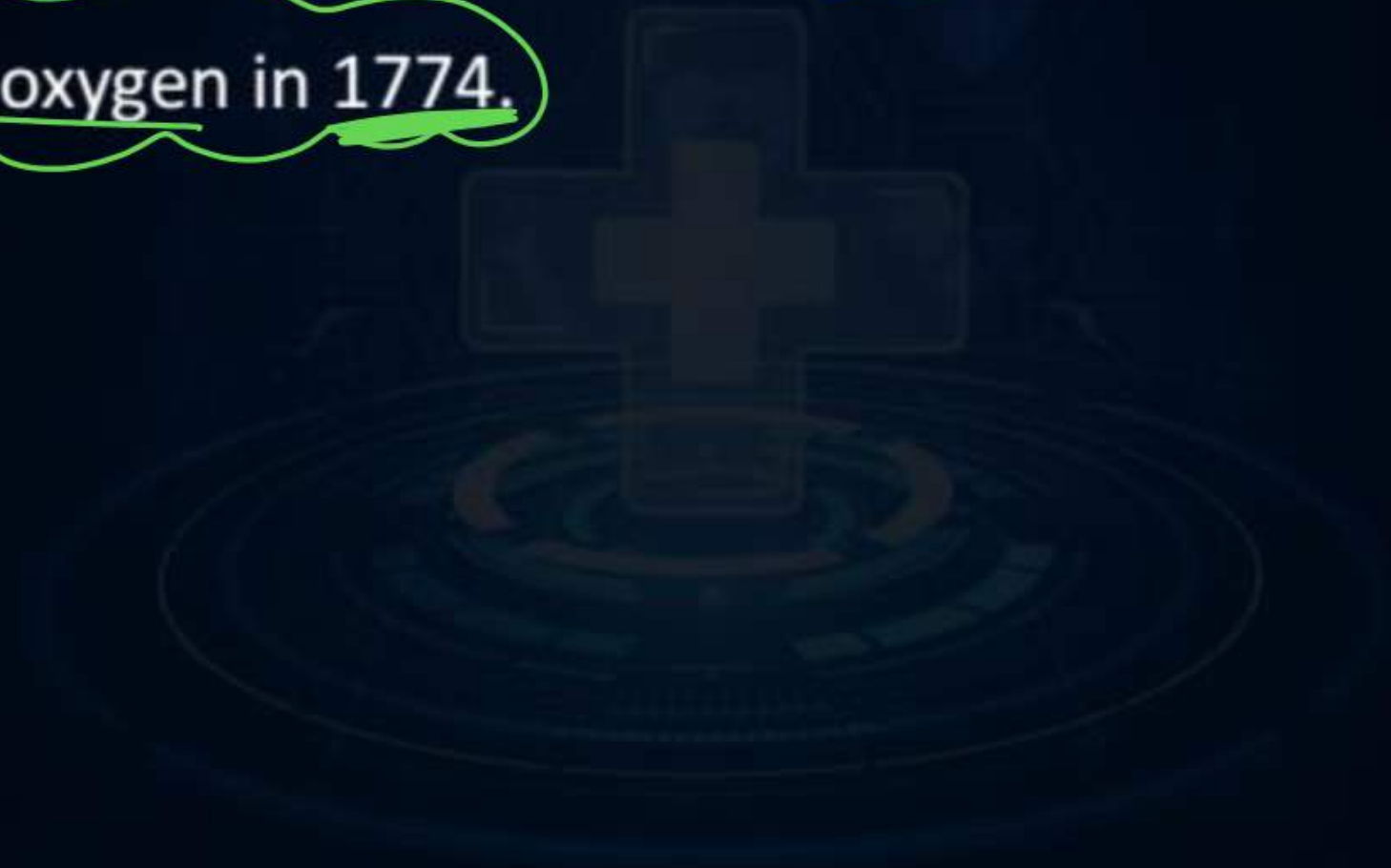
Photosynthesis in Higher Plants

24

Joseph Priestley (1733-1804) in 1770 performed a series of experiments that revealed the essential role of air in the growth of green plants.

Bell-jar exp.

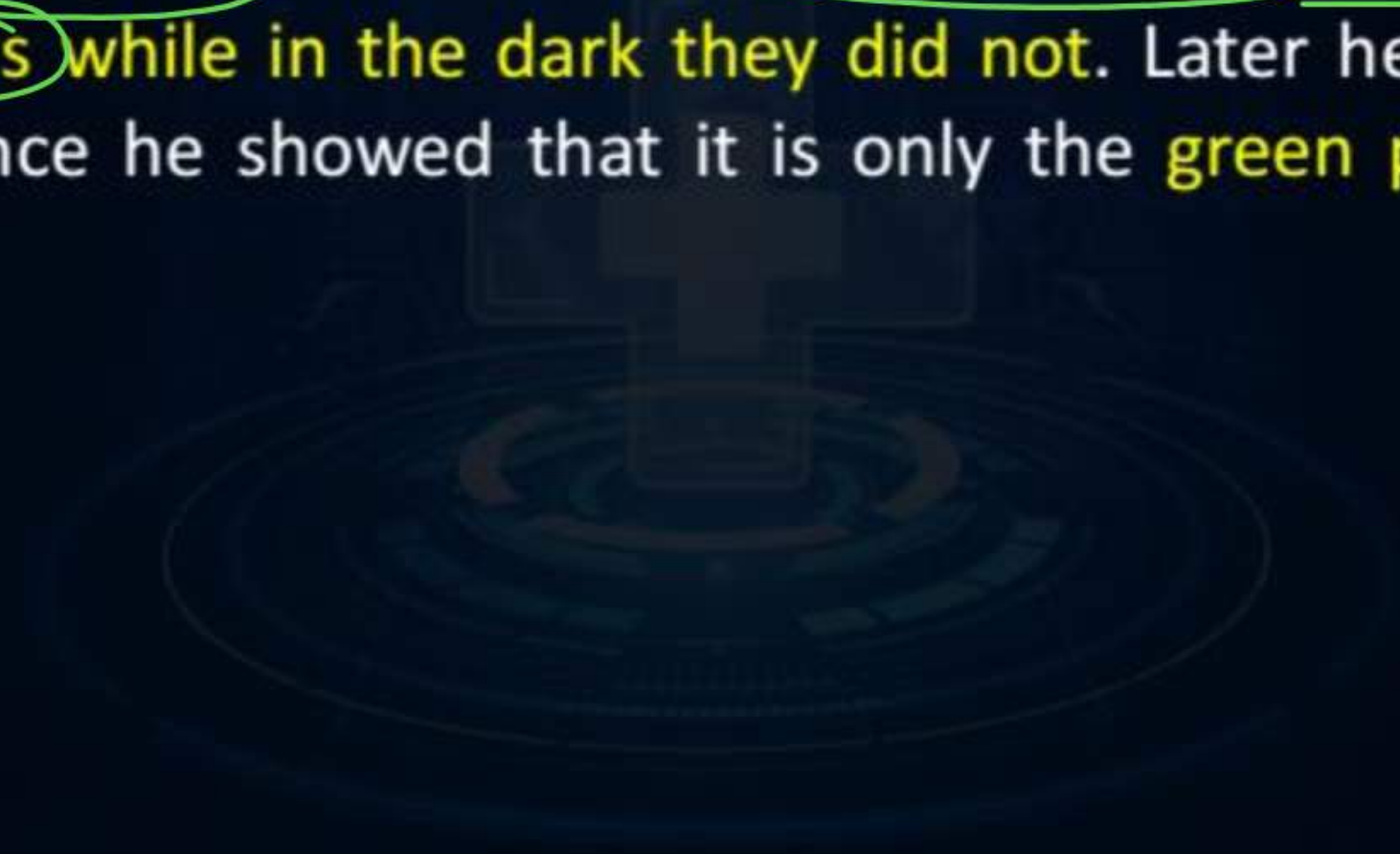
Priestley, you may recall, discovered oxygen in 1774.





25

Jan Ingenhousz (1730-1799) showed that **sunlight is essential** to the plant process that somehow purifies the air fouled by burning candles or breathing animals. Ingenhousz in an elegant experiment with an aquatic plant showed that in bright sunlight, **small bubbles were formed around the green parts** while in the dark they did not. Later he identified these bubbles to be of oxygen. Hence he showed that it is only the **green part of the plants** that could release oxygen.





(26)

~~Plant~~ → Sugar
Glucose → Store
Starch → Sp. bodies
Chloroplast

It was not until about 1854 that Julius von Sachs provided evidence for production of glucose when plants grow. Glucose is usually stored as starch. His later studies showed that the green substance in plants (chlorophyll as we know it now) is located in special bodies (later called chloroplasts) within plant cells. He found that the green parts in plants is where glucose is made, and that the glucose is usually stored as starch.



27

Now consider the interesting experiments done by T.W. Engelmann (1843 – 1909).

Using a prism he split light into its spectral components and then illuminated a green alga, Cladophora, placed in a suspension of aerobic bacteria. The bacteria were used to detect the sites of O_2 evolution. He observed that the bacteria accumulated mainly in the region of blue and red light of the split spectrum. A first action spectrum of photosynthesis was thus described. It resembles roughly the absorption spectra of chlorophyll a and b (discussed in section 13.4).

Prism
Engelmann



28

नीला (पानी)

→ O_2 comes from H_2O

Cornelius van Niel (1897-1985), who, based on his studies of purple and green bacteria, demonstrated that photosynthesis is essentially a light-dependent reaction in which hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrates.





29
Chemiosmotic Hypothesis : Peter Mitchell

30
Melvin Calvin : The use of radioactive ^{14}C by him in algal photosynthesis studies led to the discovery that the first CO_2 fixation product was a 3-carbon organic acid. He also contributed to working out the complete biosynthetic pathway; hence it was called Calvin cycle after him.

31
 C_4 Pathway : by Hatch & Slack pathway

32
Blackman's (1905) Law of Limiting Factors comes into effect.



Respiration in Plants

33

The scheme of glycolysis was given by Gustav Embden, Otto Meyerhof, and J. Parnas, and is often referred to as the EMP pathway.

Tricarboxylic acid cycle, more commonly called as Krebs' cycle after the scientist Hans Krebs who first elucidated it.

34

Plant growth and Development





35

All this started with the observation of Charles Darwin and his son Francis Darwin when they observed that the coleoptiles of canary grass responded to unilateral illumination by growing towards the light source (phototropism). After a series of experiments, it was concluded that the tip of coleoptile was the site of transmittable influence that caused the bending of the entire coleoptile.

36

Auxin was isolated by F.W. Went from tips of coleoptiles of oat seedlings.



The 'bakanae' (foolish seedling) disease of rice seedlings, was caused by a fungal pathogen Gibberella fujikuroi.

³⁷
E. Kurosawa (1926) reported the appearance of symptoms of the disease in rice seedlings when they were treated with sterile filtrates of the fungus. The active substances were later identified as gibberellic acid.

³⁸
F. Skoog and his co-workers observed that from internodal segments of tobacco stems callus (a mass of undifferentiated cells) proliferated only if, in addition to auxins, the nutrients medium was supplemented with one of following: extracts of vascular tissues, yeast extract, coconut milk or DNA.



39

Miller et al. (1955), later identified and crystallised cytokinesis promoting active substance that they termed kinetin.

40

During mid-1960s, three independent researches reported the purification and chemical characterisation of three different kinds of inhibitors: inhibitor-B, abscission II and dormin. Later all the three were proved to be chemically identical. It was named abscisic acid (ABA).

41

→ Ethylene

H.H. Cousins (1910) confirmed release of a volatile substance from ripened oranges that hastened ripening of stored unripened bananas.



Human physiology



Alfonso Corti – cardiovascular systems of reptiles . Mammalian auditory system

42

Organ of Corti (Ear)

43 Malpighian body = Glomerulus
(Renal Corpuscle) + Bowman's Capsule

44 Anton Von Leeuwenhoek : 1st living cell

Breathing and exchange of gases

Body fluids and circulation

Excretory products and their elimination

Locomotion and movement





NEURAL CONTROL AND COORDINATION

The cell body contains cytoplasm with typical cell organelles and certain granular bodies called **Nissl's granules**.

(45)

RER + ribosomes → Protein Synthesis

The gaps between two adjacent myelin sheaths are called **nodes of Ranvier**.

(46)





Chemical coordination and Integration

Adrenal Gland

47

Underproduction of hormones by the adrenal cortex alters carbohydrate metabolism causing acute weakness and fatigue leading to a disease called **Addison's disease.**

↓ ↓ deficiency
Glucocorticoids
+
Mineralocort.



Sexual Reproduction in Flowering Plants

Panchanan Maheshwari

48

Worked on embryological aspects and popularized use of embryological characters in taxonomy.

He established Department of Botany , University of Delhi.

Worked on test tube fertilization and intra-ovarian pollination

Bringing out the very first text books of Biology for Higher Secondary Schools by NCERT in 1964.

Human reproduction





Reproductive health:

Family planning⁽⁴⁹⁾ were initiated in 1951

The Medical Termination of Pregnancy (Amendment) Act, 2017⁽⁵¹⁾ was enacted by the government of India with the intension of reducing incidence of illegal abortion and consequent maternal mortality and morbidity

Government of India legalised MTP in 1971⁽⁵⁰⁾ with some strict conditions to avoid its misuse

PRINCIPLES OF INHERITANCE AND VARIATIONS





52

Gregor Mendel, conducted hybridisation experiments on garden peas for seven years (1856-1863) and proposed the laws of inheritance

Mendel proposed that something was being stably passed down, unchanged, from parent to offspring through the gametes, over successive generations. He called these things as 'factors'. Now we call them as genes.

Mendel : Test cross



Based on his observations on monohybrid crosses **Mendel** proposed two general rules to consolidate his understanding of inheritance in monohybrid crosses. Today these rules are called the Principles or Laws of Inheritance: the First Law or Law of Dominance and the Second Law or Law of Segregation And Law of Independent Assortment

Mendel published his work on inheritance of characters in **1865** but for several reasons, it remained unrecognised till 1900.

✱



53

In 1900, three Scientists (de Vries, Correns and von Tschermak) independently rediscovered Mendel's results on the inheritance of characters.

By 1902, the chromosome movement during meiosis had been worked out.

54



55

Sutton and Boveri argued that the pairing and separation of a pair of chromosomes would lead to the segregation of a pair of factors they carried. Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it the chromosomal theory of inheritance (1902)

Following this synthesis of ideas, experimental verification of the chromosomal theory of inheritance by Thomas Hunt Morgan and his colleagues, led to discovering the basis for the variation that sexual reproduction produced.

56

→ Father of Experimental Genetics



Morgan attributed this due to the physical association or linkage of the two genes and coined the term linkage to describe this physical association of genes on a chromosome and term recombination to describe generation of non-parental gene combinations





57

His student Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome.

1st linkage map
Gene



★ 58

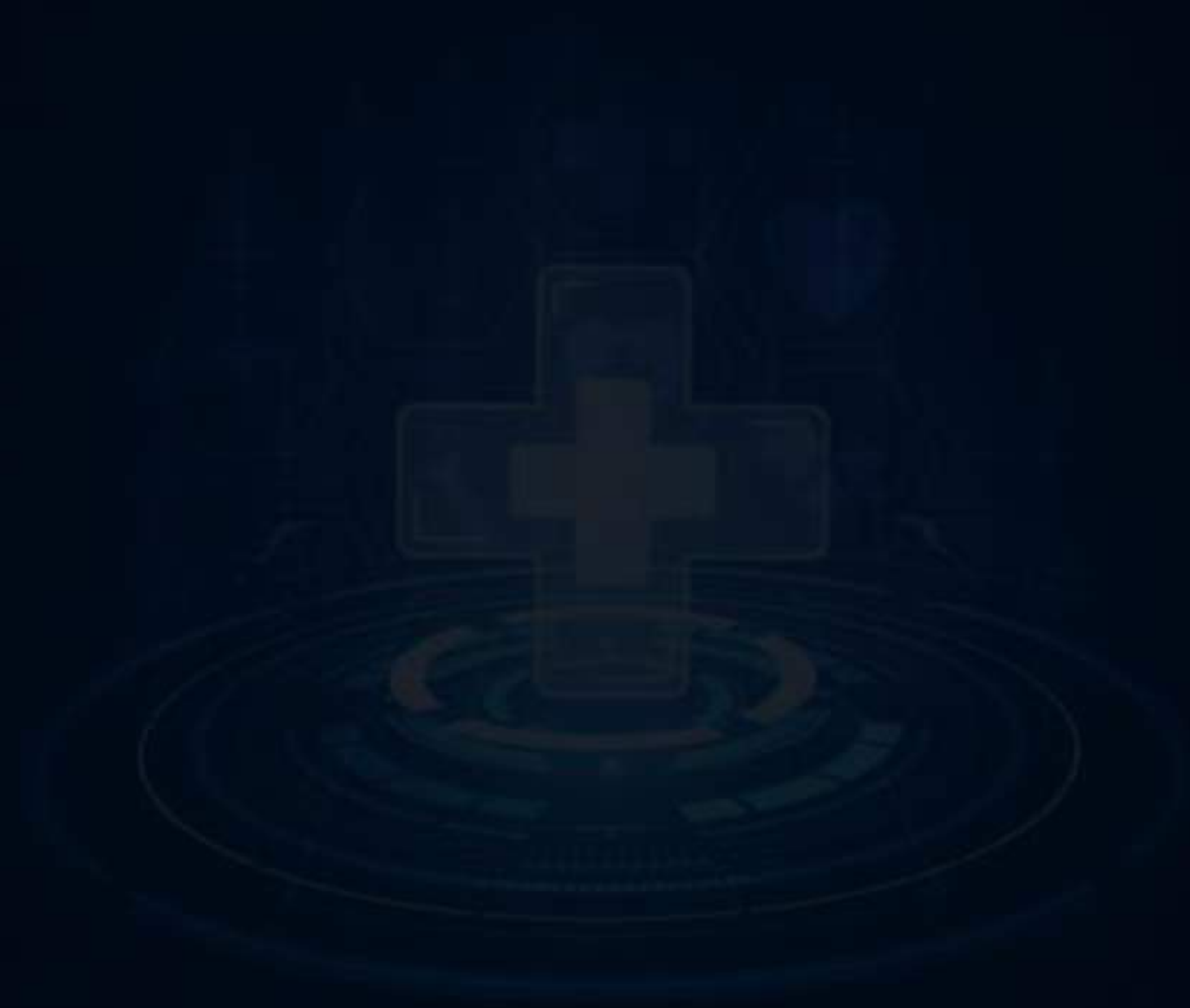
Henking (1891) could trace a specific nuclear structure all through **spermatogenesis** in a few insects, and it was also observed by him that **50 per cent of the sperm** received this structure after spermatogenesis, whereas the **other 50 per cent sperm** did not receive it. Henking gave a name to this structure as the X body but he could not explain its significance. Further investigations by other scientists led to conclusion that **'X body' of Henking was in fact a chromosome** and that is why it was given the name X-chromosome.



59

Down's Syndrome : first described by Langdon Down (1866).

MOLECULAR BASIS OF INHERITANCE





DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869. He named it as 'Nuclein'. 60

It was only in 1953 that James Watson and Francis Crick, based on the X-ray diffraction data produced by Maurice Wilkins and Rosalind Franklin, proposed a very simple but famous Double Helix model for the structure of DNA. 62



One of the hallmarks of their proposition was base pairing between the two strands of polynucleotide chains. However, this proposition was also based on the observation of Erwin Chargaff that for a double stranded DNA, the ratios between Adenine and Thymine and Guanine and Cytosine are constant and equals one.

(62)

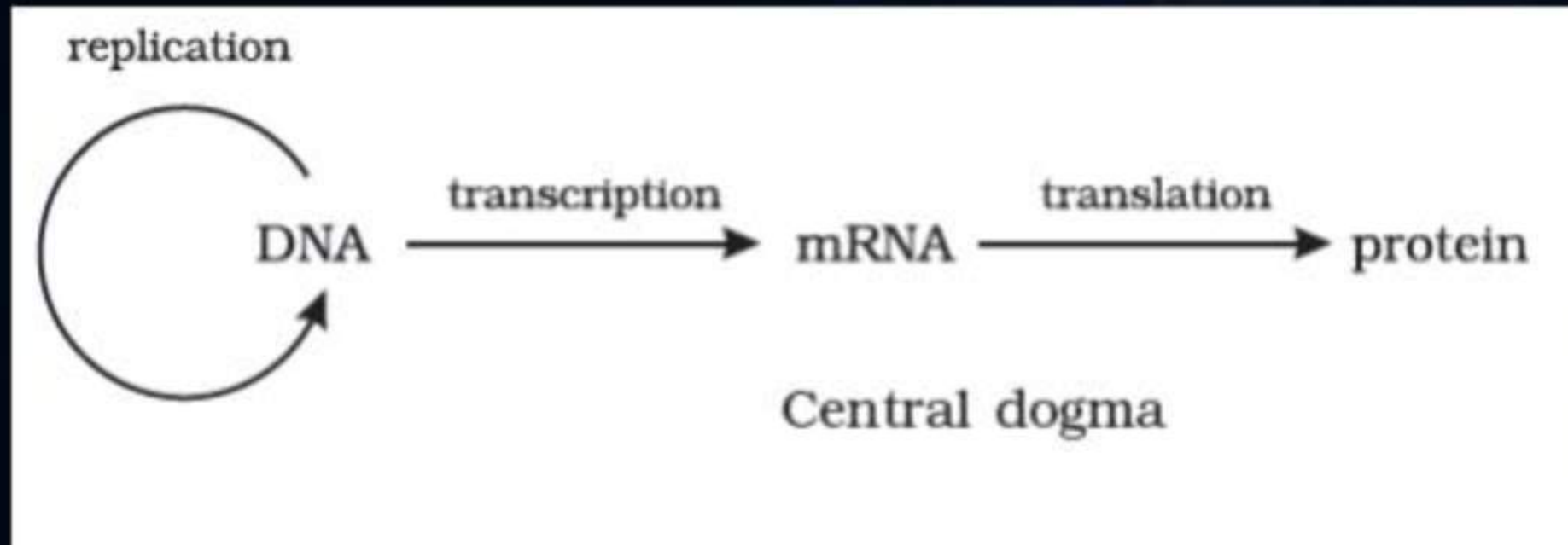
$$P_u = P_y$$
$$A + G = T + C$$

$$\frac{P_u}{P_y} = 1$$



63

Francis Crick proposed the Central dogma in molecular biology, which states that the genetic information flows from DNA-RNA-Protein.





Transforming Principle

(64)

In 1928, Frederick Griffith, in a series of experiments with Streptococcus pneumoniae (bacterium responsible for pneumonia), witnessed a miraculous transformation in the bacteria.





Biochemical Characterisation of Transforming Principle

Prior to the work of Oswald Avery, Colin MacLeod and Maclyn McCarty (1933-44), the genetic material was thought to be a protein. They worked to determine the biochemical nature of 'transforming principle' in Griffith's experiment.



66

The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase (1952). They worked with viruses that infect bacteria called bacteriophages.





It is now proven that **DNA replicates semiconservatively.** It was shown first in **Escherichia coli** and subsequently in higher organisms, such as plants and human cells. **Matthew Meselson and Franklin Stahl** performed following experiment in 1958

(67)



Very similar experiments involving use of radioactive thymidine to detect distribution of newly synthesised DNA in the chromosomes was performed on *Vicia faba* (faba beans) by Taylor and colleagues in 1958. The experiments proved that DNA in chromosomes also replicate semiconservatively.

68

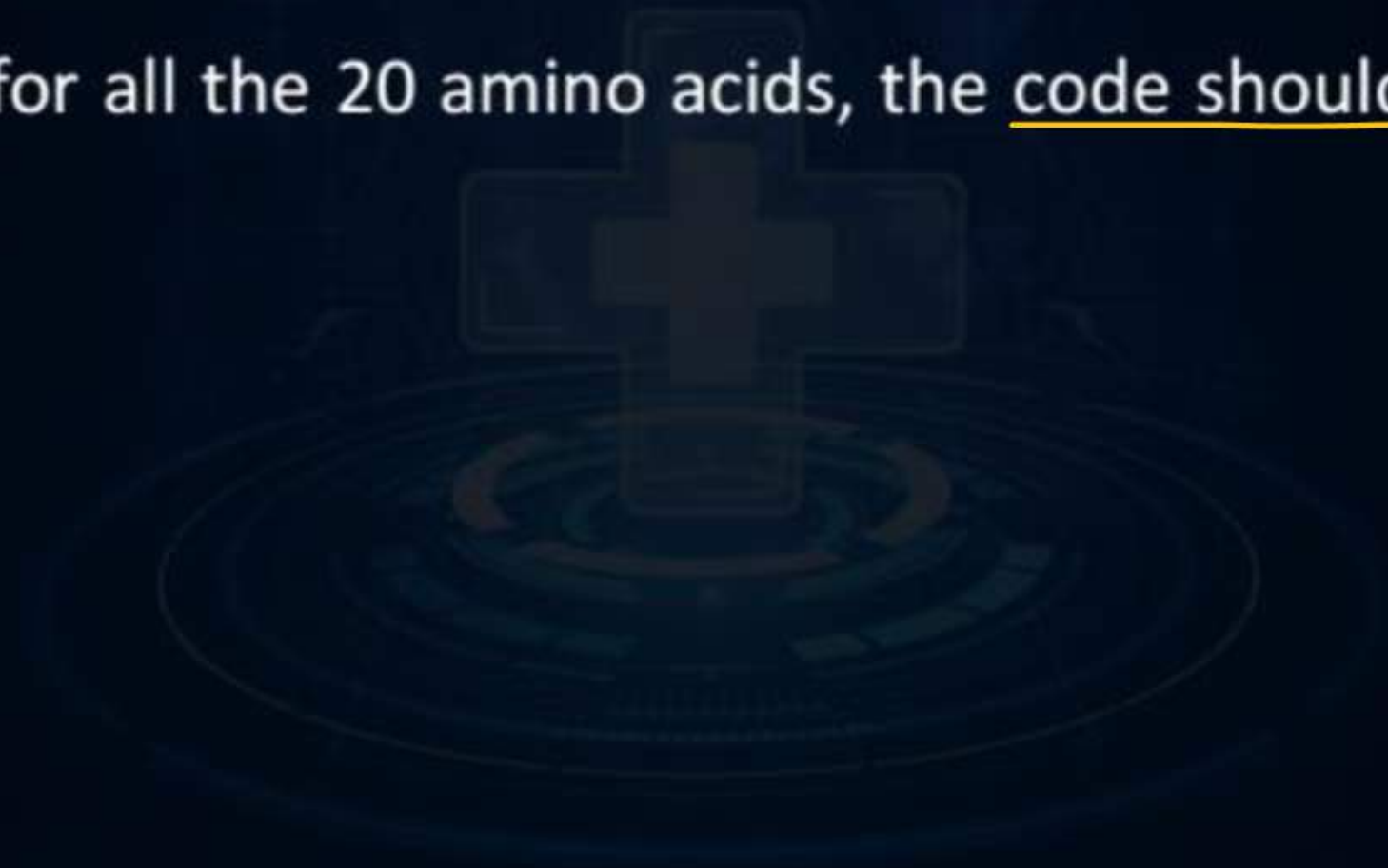


GENETIC CODE

(69)

It was George Gamow, a physicist, who argued that since there are only 4 bases and if they have to code for 20 amino acids, the code should constitute a combination of bases.

He suggested that in order to code for all the 20 amino acids, the code should be made up of three nucleotides





70

Providing proof that the codon was a triplet, was a more daunting task. The chemical method developed by **Har Gobind Khorana** was instrumental in synthesising RNA molecules with defined combinations of bases (homopolymers and copolymers).

71

Marshall Nirenberg's cell-free system for protein synthesis finally helped the code to be deciphered.

72

Severo Ochoa enzyme (polynucleotide phosphorylase) was also helpful in polymerising RNA with defined sequences in a template independent manner (enzymatic synthesis of RNA).



The elucidation of the **lac operon** was also a result of a close association between a geneticist, **Francois Jacob** and a biochemist, **Jacque Monod**.

73



74

HUMAN GENOME PROJECT: launched in the year 1990.

The Human Genome Project was a 13-year project coordinated by the **U.S. Department of Energy and the National Institute of Health**.

During the early years of the HGP, the **Wellcome Trust (U.K.)** became a major partner; additional contributions came from Japan, France, Germany, China and others.

75

The project was **completed in 2003.**



The fragments were sequenced using automated DNA sequencers that worked on the principle of a method developed by Frederick Sanger. (Remember, Sanger is also credited for developing method for determination of amino acid sequences in proteins).

The sequence of chromosome 1 was completed only in May 2006 (this was the last of the 24 human chromosomes – 22 autosomes and X and Y – to be sequenced).

The technique of DNA Fingerprinting was initially developed by Alec Jeffreys.

Evolution

79

The universe is very old – almost 20 billion years old. Big bang theory





80
Earth was supposed to have been formed about 4.5 billion years back.

81
Life appeared 500 million years after the formation of earth, i.e., almost four billion years back.

82
Louis Pasteur by careful experimentation demonstrated that life comes only from pre-existing life. He showed that in pre-sterilised flasks, life did not come from killed yeast while in another flask open to air, new living organisms arose from 'killed yeast'. Spontaneous generation theory was dismissed once and for all.



OR

83

Oparin of Russia and Haldane of England proposed that the first form of life could have come from pre-existing non-living organic molecules (e.g. RNA, protein, etc.) and that formation of life was preceded by chemical evolution, i.e., formation of diverse organic molecules from inorganic constituents.

84

In 1953, S.L. Miller, an American scientist created similar conditions in a laboratory scale (Figure 7.1). He created electric discharge in a closed flask containing CH_4 , H_2 , NH_3 and water vapour at ~~8000°C~~. He observed formation of amino acids.

800°C



85

The first non-cellular forms of life could have originated 3 billion years back. They would have been giant molecules (RNA, Protein, Polysaccharides, etc.). These capsules reproduced their molecules perhaps.

86

The first cellular form of life did not possibly originate till about 2000 million years ago. These were probably single-cells.



87

Conventional religious literature tells us about the theory of special creation. This theory has three connotations. One, that all living organisms (species or types) that we see today were created as such. Two, that the diversity was always the same since creation and will be the same in future also. Three, that earth is about 4000 years old.

Sea voyage in a sail ship called H.M.S. Beagle round the world, Charles Darwin

88



The fitness, according to ~~Darwin~~, refers ultimately and only to reproductive fitness. Hence, those who are better fit in an environment, leave more progeny than others. These, therefore, will survive more and hence are selected by nature. He called it natural selection and implied it as a mechanism of evolution.

Let us also remember that Alfred Wallace⁸⁹, a naturalist who worked in Malay Archipelago had also come to similar conclusions around the same time.



90
Embryological support for evolution was also proposed by Ernst Haeckel based upon the observation of certain features during embryonic stage common to all vertebrates that are absent in adult. For example, the embryos of all vertebrates including human develop a row of vestigial gill slit just behind the head but it is a functional organ only in fish and not found in any other adult vertebrates.

91
However, this proposal was disapproved on careful study performed by Karl Ernst von Baer. He noted that embryos never pass through the adult stages of other animals.



92

Another interesting observation supporting evolution by natural selection comes from England. In a collection of moths made in 1850s, i.e., before industrialisation set in, it was observed that there were more white-winged moths on trees than dark-winged or melanised moths. However, in the collection carried out from the same area, but after industrialisation, i.e., in 1920, there were more dark-winged moths in the same area, i.e., the proportion was reversed.

93

During his journey Darwin went to Galapagos Islands.



94

Branching descent and natural selection are the two key concepts of Darwinian Theory of Evolution (Figures 7.7 and 7.8).

Even before Darwin, a French naturalist Lamarck had said that evolution of life forms had occurred but driven by use and disuse of organs. He gave the examples of Giraffes

It is possible that the work of Thomas Malthus on populations influenced Darwin.

96



MECHANISM OF EVOLUTION

In the first decade of twentieth century, Hugo deVries based on his work on evening primrose brought forth the idea of mutations – large difference arising suddenly in a population. He believed that it is mutation which causes evolution and not the minor variations (heritable) that Darwin talked about.



98

HARDY-WEINBERG PRINCIPLE

This principle says that allele frequencies in a population are stable and is constant from generation to generation. The gene pool (total genes and their alleles in a population) remains a constant. This is called genetic equilibrium. Sum total of all the allelic frequencies is 1.

A BRIEF ACCOUNT OF EVOLUTION

About 2000 million years ago (mya) the first cellular forms of life appeared on earth.



By the time of 500 mya, invertebrates were formed and active.

Jawless fish probably evolved around 350 mya.

Sea weeds and few plants existed probably around 320 mya.

Fish with stout and strong fins could move on land and go back to water. This was about 350 mya. In 1938, a fish caught in South Africa happened to be a Coelacanth which was thought to be extinct.



In the next 200 millions years or so, reptiles of different shapes and sizes dominated on earth.

Some of these land reptiles went back into water to evolve into fish like reptiles probably 200 mya (e.g. Ichthyosaurs).

About 65 mya, the dinosaurs suddenly disappeared from the earth.



ORIGIN AND EVOLUTION OF MAN

About 15 mya, primates called Dryopithecus and Ramapithecus were existing.

Few fossils of man-like bones have been discovered in Ethiopia and Tanzania. These revealed hominid features leading to the belief that about 3-4 mya, man-like primates walked in eastern Africa. They were probably not taller than 4 feet but walked up right.



Two mya, Australopithecines probably lived in East African grasslands. Evidence shows they hunted with stone weapons but essentially ate fruit.

This creature was called the first human-like being the hominid and was called Homo habilis. The brain capacities were between 650-800cc. They probably did not eat meat.

Fossils discovered in Java in 1891 revealed the next stage, i.e., Homo erectus about 1.5 mya. Homo erectus had a large brain around 900cc. Homo erectus probably ate meat.



The Neanderthal man with a brain size of 1400cc lived in near east and central Asia between 1,00,000-40,000 years back. They used hides to protect body and buried their dead. ✓

During ice age between 75,000-10,000 years ago modern Homo sapiens arose. ✓

Pre-historic cave art developed about 18,000 years ago at Bhimbetka rock shelter, Raisen district of Madhya Pradesh. ✓

Agriculture came around 10,000 years back and human settlements started. ✓

- 1) Universe → 20 BYA
- 2) Earth → 4.5 BYA
- 3) Life appeared on Earth — after 500 million yr — 4 BYA
- 4) 1st Non-cellular life form — 3 BYA
- 5) 1st Cellular " " — 2 BYA (2000 MYA)
- 6) Invertebrates — 500 MYA
- 7) Jawless fishes — 350 MYA
- 8) Fish with stout & strong fins — 350 MYA
- 9) Sea weeds & few plants — 320 MYA
- 10) Coclocanth — South Africa — 1938
- 11) Fish-like Reptiles (eg. Ichthyosaurs) — 200 MYA
- 12) Dinosaurs extinct — 65 MYA

- 13) Dryopithecus — 15 Mya
- 14) Ramapithecus — East African grass — 2 mya
- 15) Australopithecus — 2 mya
- 16) Homo habilis — 650-800 cc
- 17) Homo erectus — 1891 — 900 cc (1.5 my)
- 18) Neanderthal man — 1 lac to 40,000 yr — 1400 cc, East & Central Asia
- 19) Ice age — 75000-10,000 yr. ago
- 20) Modern H. sapiens arose — yr. ago
- 21) Cave art — 18,000 yr. ago
- 22) Agriculture — 10,000 yr. ago
- 23) Aquatic Mammals — Whale, Dolphins, Seals, Sea Cow

Human health and disease

116

→ father of Indian Green revolution.

117

Monkambu Sambasivan Swaminathan initiated collaboration with Norman Borlaug, Green revolution, introduction of Mexican varieties of Wheat in India.

Health, for a long time, was considered as a state of body and mind where there was a balance of certain 'humors'.

117

This is what early Greeks like Hippocrates as well as Indian Ayurveda system of medicine asserted. It was thought that persons with 'blackbile' belonged to hot personality and would have fevers. This idea was arrived at by pure reflective thought. The discovery of blood circulation by William Harvey using experimental method and the demonstration of normal body temperature in persons with blackbile using thermometer disproved the 'good humor' hypothesis of health.

118



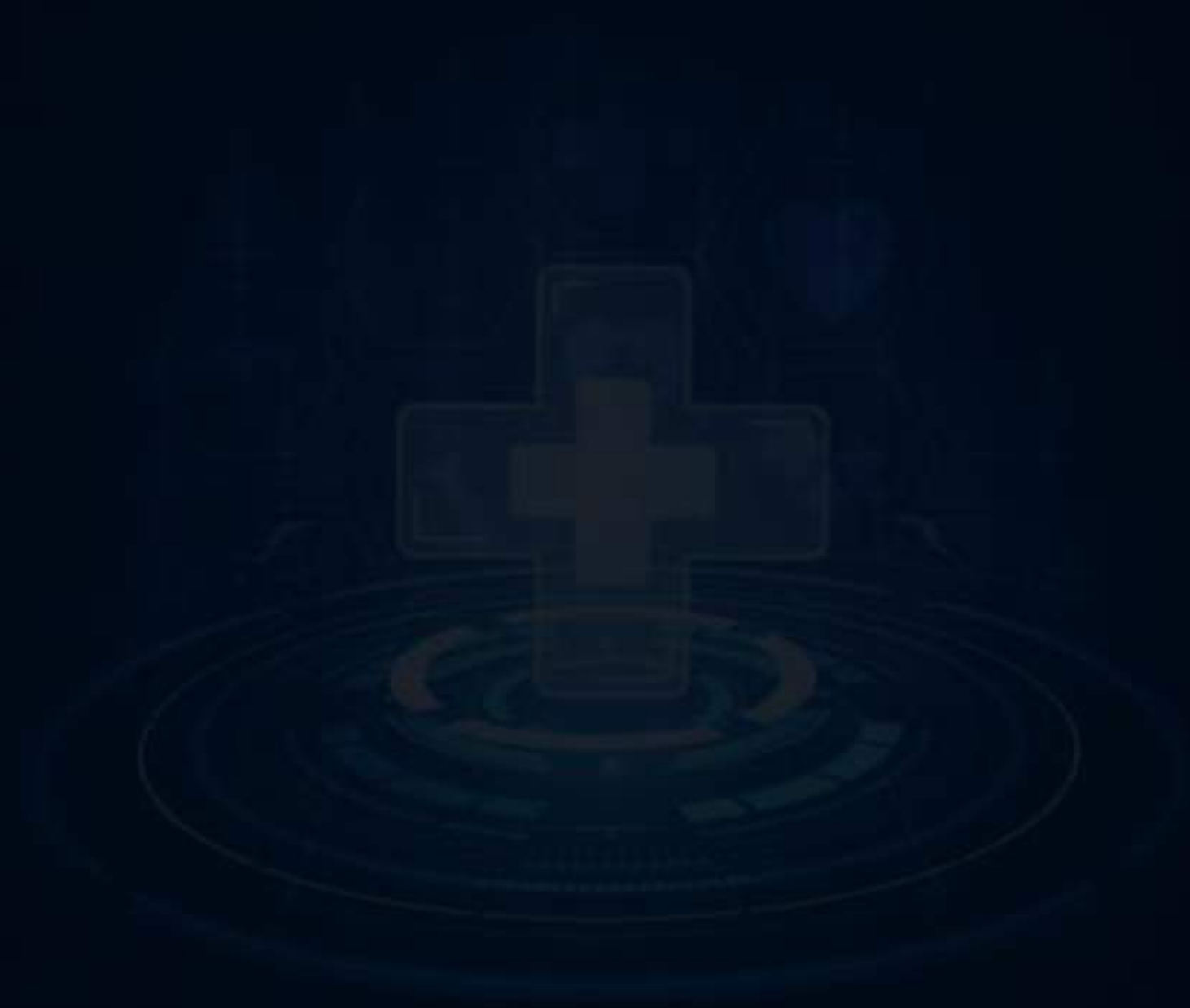
119
A classic case in medicine, that of Mary Mallon nicknamed Typhoid Mary, is worth mentioning here.

There is always a time-lag between the infection and appearance of AIDS symptoms. This period may vary from a few months to many years (usually 5-10 years). 120

Smoking also paves the way to hard drugs. Tobacco has been used by human beings for more than 400 years. 121

122
The period between 12-18 years of age may be thought of as adolescence period.

Microbes in Human Welfare





Do you know that Penicillin was the first antibiotic to be discovered, and it was a chance discovery? Alexander Fleming while working on Staphylococci bacteria.

123

However, its full potential as an effective antibiotic was established much later by Ernest Chain and Howard Florey.

124

This antibiotic was extensively used to treat American soldiers wounded in World War II. Fleming, Chain and Florey were awarded the Nobel Prize in 1945, for this discovery.

125

Biotechnology: Principles and Processes

The construction of the first recombinant DNA Stanley Cohen and Herbert Boyer accomplished this in 1972

126



Restriction Enzymes

127

In the year 1963, the two enzymes responsible for restricting the growth of bacteriophage in Escherichia coli were isolated. One of these added methyl groups to DNA, while the other cut DNA. The later was called restriction endonuclease.

The first restriction endonuclease—Hind II, whose functioning depended on a specific DNA nucleotide sequence was isolated and characterised five years later.

128

today we know more than 900 restriction enzymes that have been isolated from over 230 strains of bacteria each of which recognise different recognition sequences.

129

Biotechnology and its Applications





130

At present, about 30 recombinant therapeutics have been approved for human-use the world over. In India, 12 of these are presently being marketed.

132

131

In 1983, Eli Lilly an American company prepared two DNA sequences corresponding to A and B, chains of human insulin and introduced them in plasmids of E. coli to produce insulin chains.

133

The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency.



In ¹³⁴1997, the first transgenic cow, ¹³⁵Rosie, produced human protein-enriched milk (2.4 grams per litre). The milk contained the human alpha-lactalbumin and was nutritionally a more balanced product for human babies than natural cow-milk.

There are an estimated ¹³⁵200,000 varieties of rice in India alone.

¹³⁶Basmati rice is distinct for its unique aroma and flavour and 27 documented varieties of Basmati are grown in India.

In ¹³⁷1997, an American company got patent rights on Basmati rice through the US Patent and Trademark Office.

Organisms and Populations





138

Ramdeo Misra (1908-1998) – Father of Ecology in India





In 1981 the r value for human population in India was 0.0205

Logistic growth: sigmoid curve. This type of population growth is called Verhulst-Pearl Logistic Growth

Life History Variation

Populations evolve to maximise their reproductive fitness, also called Darwinian fitness (high r value), in the habitat in which they live.

Predation : The prickly pear cactus introduced into Australia in the early 1920's caused havoc



143

(ii) Competition: When Darwin spoke of the struggle for existence and survival of the fittest in nature, he was convinced that interspecific competition is a potent force in organic evolution.

144

It is relatively easy to demonstrate in laboratory experiments, as Gause and other experimental ecologists did, when resources are limited the competitively superior species will eventually eliminate the other species

Gause Exclusion Principle



Another evidence for the occurrence of competition in nature comes from what is called 'competitive release'.

145

Connell's elegant field experiments showed that on the **rocky sea coasts of Scotland**, the **larger and competitively superior barnacle Balanus dominates** the intertidal area, and excludes the smaller barnacle Chathamalus from that zone.

छोटा
Chota

↓
बड़ा
Bada



146

Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

Species facing competition might evolve mechanisms that promote co-existence rather than exclusion. One such mechanism is 'resource partitioning'.

147

MacArthur showed five closely related species of warblers living on the same tree were able to avoid competition and co-exist due to behavioural differences in their foraging activities

Ecosystem

148

10% Law – by Lindmann (1942), follow Second law of Thermodynamics



Biodiversity and Conservation





Biodiversity is the term popularized by the sociobiologist Edward Wilson to describe the combined diversity at all the levels of biological organisation.

According to the International Union for Conservation of Nature and Natural Resources (IUCN) (2004), the total number of plant and animal species described so far is slightly more than 1.5 million.

Some extreme estimates range from 20 to 50 million, but a more conservative and scientifically sound estimate made by Robert May places global species diversity at about 7 million.



Species-Area relationships:

During his pioneering and extensive explorations in the wilderness of **South American jungles**, the great **German naturalist and geographer Alexander von Humboldt** observed that within a region species richness increased with increasing explored area, but only up to a limit.





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We don't know how these attributes are linked to species richness in a community, but David Tilman's long-term ecosystem experiments using outdoor plots provide some tentative answers. Tilman found that plots with more species showed less year-to-year variation in total biomass. He also showed that in his experiments, increased diversity contributed to higher productivity.



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Analogy (the 'rivet popper hypothesis') used by Stanford ecologist Paul Ehrlich.

❖ Effect of species loss to ecosystem health-SPEAR

- S-Stanford Airplane=Ecosystem
- P-Paul Rivets=Species
- E- Ehrlich Rivets on wings=Key species
- A-Airplane analogy
- R-Rivet popper hypothesis

Key species = that drive major ecosystem functions



The IUCN Red List (2004) documents the extinction of 784 species (including 338 vertebrates, 359 invertebrates and 87 plants) in the last 500 years.

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Invertebrates > Vertebrates > Plants

359

338

87

359

(359 - 24)

During the long period (> 3 billion years) since the origin and diversification of life on earth there were five episodes of mass extinction of species.

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Ecologists warn that if present trends continue, nearly half of all species on earth might be wiped out within next 100 years.



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Over-exploitation: Many species extinctions in the last 500 years (Steller's sea cow, passenger pigeon) were due to overexploitation by humans.



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The historic Convention on Biological Diversity ('The Earth Summit') held in Rio de Janeiro in 1992, called upon all nations to take appropriate measures for conservation of biodiversity and sustainable utilisation of its benefits.

159

In a follow-up, the World Summit on Sustainable Development held in 2002 in Johannesburg, South Africa, 190 countries pledged their commitment to achieve by 2010, a significant reduction in the current rate of biodiversity loss at global, regional and local levels.

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* Sat. 20/4 5pm Yakeen yt channel
Maharav. Complete Zoology

* Tues. 23/4 5pm 1000⁺ Comp. Biology

* Next Thurs. 25/4 } CW • Diag & Tables
Sat. 27/4 } 6pm • Examples

* Unmmed Series 12th Zool. CW

* PODcast CW



Thank *You*